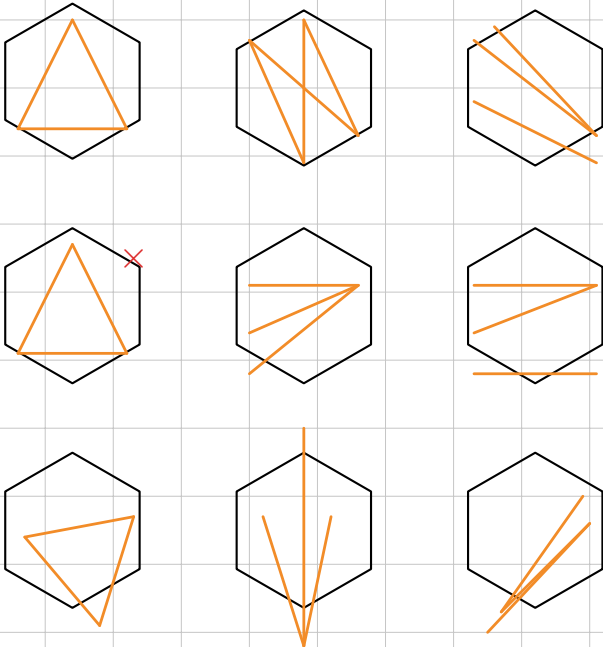
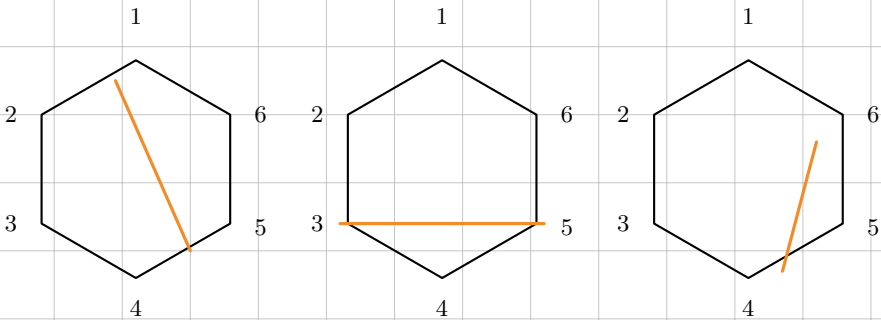


$$X_{2j} = X_{1j} - X_{13}$$

all possible diagonals with 6 legs

group them by



z -scaling depends on z scaling of propagators

boundary propagators: the ones that scale linearly with z under BCFW shift

(X_{1i} and X_{2i})

group by # of boundary propagators

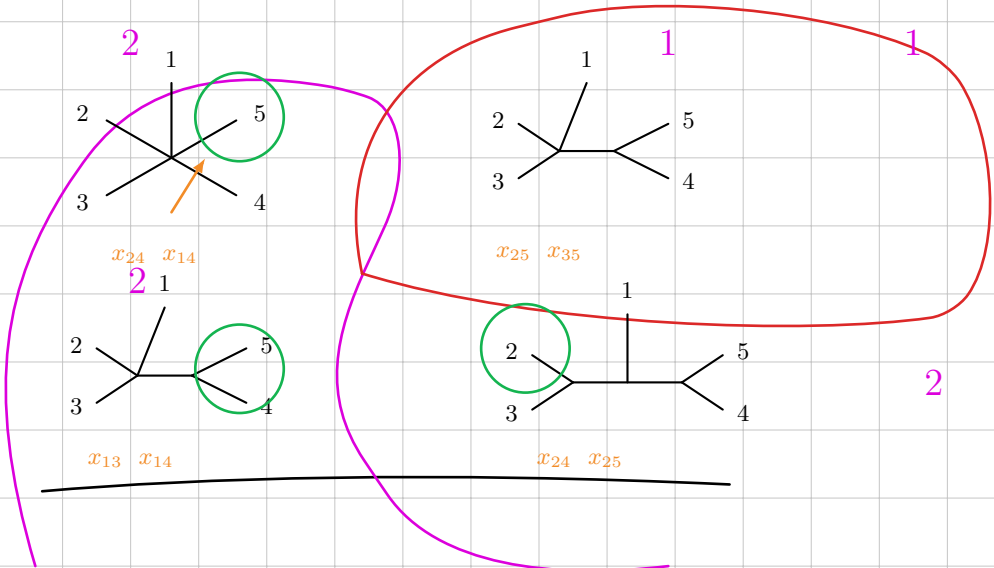
$\Rightarrow$ group them by scaling under BCFW shift

$$B = B_0 \boxed{+} B_1 + \boxed{B}_{\rightarrow 2} + \boxed{\cdots}$$

0 boundary
propagators

1 bound.
prop.

2 boundary
prop.



bound. prop. : $X_{13} \ X_{14}$
 $X_{24} \ X_{25}$

1-zero and BCFW shift

up to cyclic permutations a 1-zero is the imposition of the $n - 3$ constraints

$C_{13} = C_{14} = \cdots = C_{1,n-1} = 0$

on the kinematic data.

Since $C_{i0} = X_{i,i} + X_{i+1,i+1} - X_{i,i+1} - X_{i+1,i}$

the constraints can be equivalently be written as

$$\begin{aligned} X_{13} + X_{24} - X_{14} - X_{23} &= 0 \\ \vdots \\ X_{1,n-1} + X_{2n} - X_{1n} - X_{2,n-1} &= 0 \end{aligned}$$

$$\Leftrightarrow \begin{aligned} X_{24} &= X_{14} - X_{13} \\ \vdots \\ X_{2n} &= X_{1n} - X_{13} \end{aligned}$$

so the 1-zero fully fixes X_{2j} once X_{1j} is specified, and does not

affect X_{ij} if $i,j \notin \{1,2\}$.

the corresponding BCFW shift is the change

$$\begin{cases} P_2 \rightarrow P_2 + zq \\ P_n \rightarrow P_n - zq \end{cases}$$

where z is a complex parameter and q satisfies

$$\begin{cases} q^2 = 0 \\ q \cdot P_2 = q \cdot P_n = 0 \end{cases}$$

↓ so that

$$(P_2 + zq)^2 = 0$$

$$(P_n - zq)^2 = 0$$

Let $X_{ij}(z)$ be the z -dependent planar invariant.

$X_{ij}(z)$ depends on z only if it includes exactly one of P_2, P_n in

the sum, otherwise either there is no z or they cancel out.

$\Rightarrow$ the only z -dependent ones are $X_{1i}(z) \quad i = 3, \dots, n - 1$

and $X_{2j}(z) \quad j = 4, \dots, n.$

Let X_i^∞ denote the coefficient of the largest power of z appearing

in $X_{1i}(z)$.